# Synthetic Biological Intelligence: Interfaces for In-Vitro Neuronal Learning

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## Background & Motivation

- Modern AI models are increasingly limited by their computational and hardware costs as scale grows.
- Biological neural networks perform flexible and adaptive computation with extreme energy efficiency, far beyond current artificial systems.
- We investigate how in-vitro biological neural networks can be trained to perform goal-directed learning through a closed-loop interface. We also address future concerns of scalability, which could enable radically more energy-efficient forms of AI computation.

## Theoretical Foundations: Biology vs. Artifice

**The Problem: Backpropagation**

*Biologically implausible due to:*

- **Temporal Discontinuity:** Neurons process in real-time; they cannot pause for a "backward pass."[1]
- **Weight Transport:** Synapses are unidirectional and cannot copy weights for feedback loops.
- **Non-Local Plasticity:** Updates require downstream errors unavailable to local synapses.

**The Solution: Predictive Coding**

*Grounded in the Free Energy Principle:*

- **Generative Modeling:** The brain constantly predicts sensory input rather than just reacting.
- **Error Minimization:** Learning is local, minimizing the delta between *prediction* and *sensation*.
- **Metabolic Efficiency:** High Error → Chaotic Firing (Expensive) Low Error → Homeostasis (Efficient)

## Training Methodology

We use a three-phase, closed-loop framework for training in-vitro neural cultures, inspired by predictive coding and the Free Energy Principle [4].

**1. Characterizing neural activity**

Neuronal cultures are plated on a microelectrode array (MEA) and recorded to capture spontaneous action potentials. We then generate a spatial map of **putative neuron** clusters by identifying electrodes with high firing rates and large spike amplitudes.(Figure 1.a)

**2. Mapping connectivity**

Each putative unit is stimulated with **biphasic pulse trains** (50 pulses at 2 Hz) to identify first-, second-, and third-order connectivity and to define functional connectivity layers [4]. (Figure 1.b)

**3. Configuring the neural controller and closed-loop training**

Based on connectivity, units are assigned as **encoding**, **decoding**, or **training** channels. Encoding units receive task state, decoding units report actions, and remaining units receive reward/penalty stimulation; during closed-loop training, decoded activity controls the agent and the resulting state is fed back through the encoding units. (Figure 1.c)

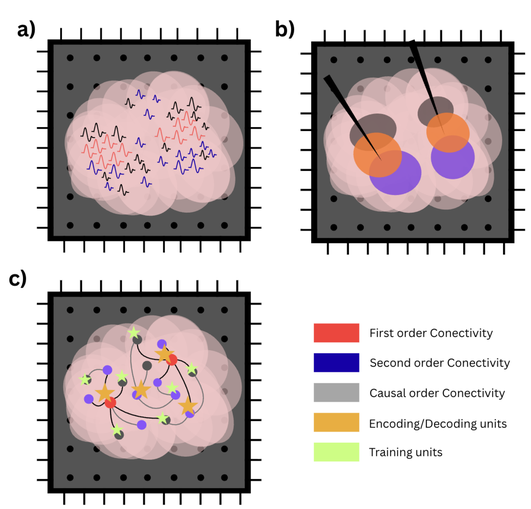


Figure 1: (a) characterize neural activity and identify putative neurons; (b) Stimulate putative units; (c) Assign roles to neurons and run closed-loop training

## Architectural Setup

Due to unrealistic layout of 2-D MEAs, which don't necessarily have layers that resemble that of a neural network, we devised a 3-D microarchitectural device which houses a multi-layered neural culture setup.

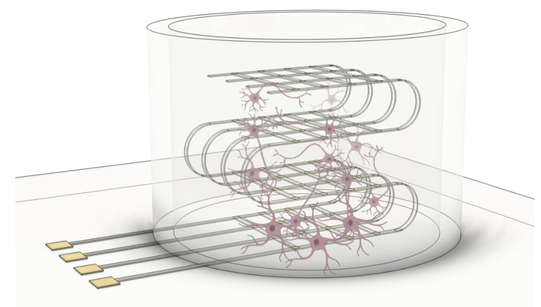

Figure 2: Graphic of 3-D Architecture

## Recording Setup

To maximize the number of recording channels in the recording setup, there is a new fabricated design to allow for 128 Channels for recording rather than just 64 Channels.

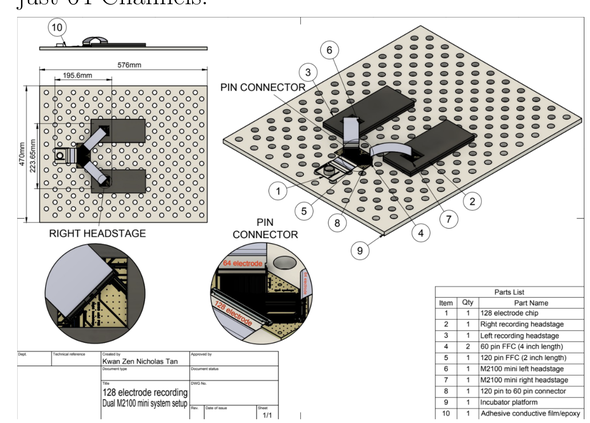


Figure 3: New MEA Recording Design

## Results and Future Concerns

We preliminarily tested Action Potentials from our setup to ensure recordings were clean and stable.

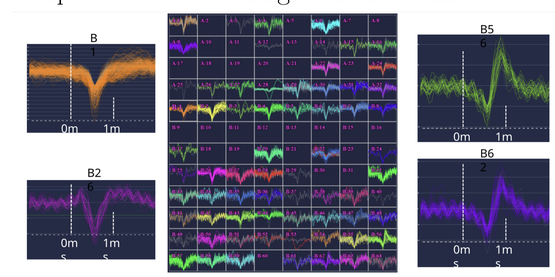


Figure 4: Action potentials recorded from our setup

Given our setup, we hope to explore 3 different tasks in the future. **1)** Define a learning tasks most suitable for our neural cultures [2-4]. **2)** Explore and measure the performance of our 3-D culture compared to our 2-D culture in terms of learning capability and efficiency. **3)** Scalability remains one of the major concerns when considering organic computers such as biological neural networks. To this end, we envision that this study can contribute significantly to: **A)**Assessing the impact of increased recording channels on the overall system's processing power. **B)**Exploring multi-culture topologies, where independent biological agents solve sub-tasks to achieve a unified computational goal.

## References & Acknowledgement

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[4] Rountree, C. et al. (2025) Long-term potentiation and closed-loop learning in paired brain organoids. bioRxiv.